

NADINE EL-MUFTI

Canadian Citizen, residing in 📍 Montreal, QC

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EXECUTIVE SUMMARY

Clinically grounded AI researcher specializing in speech-based biomarkers for neurological and psychiatric disease detection, interpretability, and AI governance. Master's candidate at Concordia University and Mila – Quebec's AI Institute with experience across the full ML lifecycle and a strong focus on ethics, fairness, and responsible technology, with growing interests spanning clinical AI modalities and the intersection of cybersecurity and AI.

WORK EXPERIENCE

The Douglas Research Centre, McGill University

Graduate Research Assistant

Montreal, QC

Jun 2025 – Present

- Driving speech-based relapse prediction research in psychotic disorders in collaboration with Dr. Lena Palaniyappan and Dr. Alban Voppel by maintaining multisite data collection using REDCap workflows and powering the Voici Speech Bank by engineering and analyzing pipelines to clean, align, and structure longitudinal speech data across diverse studies. This work enables multimodal insights into voice, symptoms, and cognition, paving the way for more predictive, compassionate mental health care.

Applied Perception Lab, Concordia University

Graduate Research Assistant

Montreal, QC

Dec 2024 – Present

- Leading research at the intersection of AI, ethics, and governance, investigating anticipatory fairness in surgical AI futures and developing empirically grounded governance approaches for healthcare AI derived from diverse stakeholder voices.

Mila - Quebec AI Institute

Collaborating Researcher

Montreal, QC

Oct 2024 – Present

- Leading novel research in Medical Machine Learning for early disease detection via speech analysis, covering the entire ML lifecycle from dataset curation to model training, evaluation, optimization, and deployment.

Concordia University

Teaching Assistant

Montreal, QC

Jan 2025 – April 2025

- Achieving consistent and fair grading for Concordia's Machine Learning course 'Comp 432' by marking labs, delivering prompt feedback, and addressing inquiries from 110 students.

Applied Perception Lab

NSERC USRA Research Assistant

Montreal, QC

May 2022 – Aug 2022

- Co-authored a conference paper for Medical Image Computing and Computer Assisted Intervention (MICCAI) 2022's Ethical and Philosophical Issues in Medical Imaging (EPIMI) workshop, presenting a comprehensive User-Centered Design approach to identify and address Ventriculostomy surgeons' workflow challenges, to improve surgical efficiency.

PUBLISHED & ON-GOING RESEARCH

- El-Mufti, N., Mackinley, M., Dzialoszynski, P., Palaniyappan, L., Voppel, A.* *Automatic Speech Recognition Performance in Psychiatric Speech: Linguistic, Clinical, and Architectural Factors*. NPP—Digital Psychiatry and Neuroscience (DPN), Nature Portfolio. (Full manuscript invited; Under Review)

- El-Mufti, N., Drouin, S., Jannin, P., Kersten-Oertel, M.* *Futures Before Failures: Design Fictions for Anticipatory Fairness in Surgical Care*. (Under Review)
- El-Mufti, N., Kersten-Oertel, M., Ravanelli, M. *Enhancing Speech-Based Parkinson's Detection with Celebs4PD: A Corpus of Natural, In-the-Wild, Longitudinal Celebrity Speech*. (In Preparation)
- Reyes, J., El-Mufti, N., Gorman, S., Xie, D., Kersten-Oertel, M. *User-Centered Design for Surgical Innovations: A Ventriculostomy Case Study*. EPIMI Workshop, MICCAI 2022, LNCS vol. 13755, 2022. https://doi.org/10.1007/978-3-031-23223-7_5

LEADERSHIP, SERVICE & ENGAGEMENT

[Arabs in Neuroscience](#) (Member, Apr 2026 - Present) · [The Levantines](#) (Member, Feb 2026 - Present) · [Association Étudiante de Mila](#) (Member, Oct 2025 – Present) · [Arbour Foundation Scholars](#) (Member, Sep 2025 – Present) · [Golden Key International Honour Society](#) (Member, Sep 2025 – Present) · [Brainhack MTL](#) | [Brainhack Global](#) (Organizer, Jul 2025 – Present) · [Graduate Students' Association of Concordia GSA](#) (Subcommittee Member, Jan 2025 – Apr 2025) · [School of Graduate Studies, Concordia University](#) (Ambassador, Sep 2024 – May 2026) · [GEMinAI](#) | [Concordia AI2](#) (Mentee, Oct 2024 – May 2025)

EDUCATION

Master of Computer Science Computer Science Concordia University / Mila - Quebec AI Institute	Montreal, Quebec Jan 2025 – Present
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Bachelor of Engineering Software Engineering GPA (3.67) With Distinction Concordia University	Montreal, Quebec Jan 2019 – May 2024
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HONORS & AWARDS

Power Corporation Of Canada Graduate Fellowship Issued by Concordia University	Jun 2026
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Arbour Foundation Master's Scholarship Issued by The Arbour Foundation	Sep 2025
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FRQNT B1X Award Competition – 92/100, Ranked 9th (Funding Cutoff: 8) Issued by the Fonds de recherche du Québec	Apr 2025
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NSERC CGS-M Award Issued by the Natural Sciences and Engineering Research Council of Canada	Apr 2024
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NSERC Undergraduate Student Research Award (USRA) Issued by the Natural Sciences and Engineering Research Council of Canada	May 2022
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LICENSES & CERTIFICATIONS

[Transatlantic Summer School](#) (UdeM x LMU, Jun 2026) · [Master AI Tools for Research](#) (WiseUp Communications, Jun 2026) · [Foundations in Responsible AI & AI Ethics](#) (Mila, May 2026) · [Bell x Mila x Buzz HPC x Kids Help Phone Hackathon](#) (Mila, Apr 2026) · [Ethics and Governance of AI for Health](#) (WHO Academy, Jan 2026) · [QcSE Program](#) (V1 Studio, Jun 2025) · [Strategic Public Communications](#) (GradProSkills, Mar 2025) · [TRAIL AI Learning Certificate](#) (Mila, Mar 2025) · [TCPS 2: CORE 2022](#) (Government of Canada, Jan 2025) · [Tech Stewardship Certification](#) (Tech Stewardship Network, Jul 2024)